BOOK II — THE MATHEMATICAL / CONSTITUTIVE FRAMEWORK

> *WORKING DRAFT — part of the GRUT working corpus; statuses per `books/CORPUS_CHARTER.md`; subject to chapter-by-chapter audit; nothing here banks.*

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Contents

II.0 · What this book is

This book presents the mathematical machinery of the responsive-vacuum framework as the record actually supports it: the influence action, the admissible kernel space, the admission gates (causality, KMS/FDT, passivity, positivity), the memory structures, the one kernel the program has actually computed at its declared scope, and the structural theorems that were earned along the way. It is written for a physicist reader who wants the constitutive mathematics as a coherent object — but the coherence must not be mistaken for derivational strength, and so two theorems that *cut against* the framework's own ambitions are treated as load-bearing members of the framework itself: the form-universality result (u1: the mathematical form is generic and confers no GRUT-specific content) and the no-pin theorem (x_no_pin: the action carries a family of kernels, an amplitude-homogeneous cone, and pins no point of it). A constitutive framework that knows exactly what its mathematics does and does not fix is the only kind this record can honestly present.

Throughout, standard-physics background (Schwinger–Keldysh contours, Feynman–Vernon influence functionals, Kramers–Kronig relations, KMS states, Mori–Zwanzig projections) is summarized for readability and is *always* distinguished from what the GRUT record itself established. The register `provenance/claims.json` (74 nodes, read-only to this corpus) is the status authority wherever it speaks.

II.1 · The influence action

The framework's heart is a single Schwinger–Keldysh influence action for the metric perturbation h on a declared background. Fields are doubled on the closed time path (h^+, h^-); in the Keldysh basis h_r = (h^++h^-)/2, h_a = h^+−h^- the quadratic action reads

```
S_IF[h_r, h_a] = ∫ (dω/2π) d³k [ h_a · K_R(ω,k²) · h_r + (i/2) h_a · N(ω,k²) · h_a ]
```

with K_R the retarded (dissipation) kernel and N the noise kernel. This is the standard open-system structure: coarse-graining any local, causal open quantum system over its environment yields precisely this form (Feynman–Vernon 1963; Caldeira–Leggett 1983; non-equilibrium Keldysh field theory).

> **STATUS: RECOVERED (generic; u1: the form confers no GRUT-specific content)** — the canonical status for the influence-functional/Feynman–Vernon form; the SHOWN content is the *borrowed* universality of the form only (source: `provenance/claims.json` node `u1_form_universality`; `S_IF.md`).

The register books the formalism at an explicit price. Node `rung1_inin_formalism` (the formalism half, split by owner ruling 2026-08-23 from the ontological bet) carries **four priced inputs** (+4): the system/bath split, the Gaussian/linear-response truncation, the background Lorentzian causal structure, and — booked 2026-08-17 after an adversarial screen — the *4d-covariant availability of the Ward-sourced gauge-orbit zero* (that K_R annihilates the full 4d gauge orbit on its retarded slot at covariant strength; the frontier-reserved Bardeen completion is exactly what this input pre-pays).

> **STATUS: ASSUMPTION (AXIOM — the program's declared entry price, four priced inputs)** — a stance, booked openly with a `laundering_ok` justification; the formalism does not imply the ontology (source: `provenance/claims.json` node `rung1_inin_formalism`, ledger_note).

The companion ontological claim — that the gravitational vacuum *is* a responsive medium with finite memory — is booked separately and is not part of the mathematics at all:

> **STATUS: ASSUMPTION (AXIOM, +1; "a STANCE, explicitly not derived")** — node `rung1_ontology_finite_memory`; its strongest form (single-pole memory) is treated in §II.5 below, where its supporting history is stated on its face.

The formal action itself is written out in `S_IF.md`, a construction document at ledger zero: it declares the field content, the S1–S7 symmetry inventory (causality, quadratic truncation, system/bath split, KMS/FDT + matrix passivity, diffeomorphism invariance with its banked Ward limitation, background homogeneity/isotropy with inherited parity-evenness, and pair symmetry/Onsager — the last flagged as inherited, not chosen, and substantive for the retarded kernel). No symmetry beyond these is declared; in particular linearized Weyl invariance is *not available* to the framework, because GRUT imports α as the trace-anomaly ratio and the anomaly is the statement that Weyl invariance is broken (`p_tt_ansatz.boundary_condition`).

II.2 · The admissible kernel space: the operator basis

What tensor structures can K_R and N carry? This was made a computation rather than an assumption (`calc/operator_basis.py`, results in `calc/RESULTS_operator_basis.md`; register node `eft_operator_basis`). In the spatial-SVT frame at fixed slicing and fixed k — the same frame in which the pure-TT ansatz and the linear-cosmology work state their claims — the isotropic, parity-even, pair-symmetric kernel space is **exactly rank 5**: {$P^{(2)}$, $P^{(1)}$, $P^{(0s)}$, $P^{(0w)}$, MIX}. Exhaustiveness is *computed, not asserted*: the instrument builds the little-group representation and finds the pair-symmetric commutant is exactly 5-dimensional (full commutant 6; the +1 is the single pair-antisymmetric trace-longitudinal mixer, itself Ward-killed). The linearized-diffeo (Ward) orbit is annihilated by $P^{(2)}$ and $P^{(0s)}$ and hit by the rest; helicity forbids TT ↔ scalar mixing. The Ward-surviving family is therefore **exactly two independent coefficient functions**:

$$K\_R(\omega,k^2) = c_2(\omega,k^2)\cdot P^{(2)} + c_0(\omega,k^2)\cdot P^{(0s)}$$
$$N(\omega,k^2) = n_2(\omega,k^2)\cdot P^{(2)} + n_0(\omega,k^2)\cdot P^{(0s)}$$

> **STATUS: DERIVED (enumerated frame/order; exhaustiveness computed via the little-group commutant, independently reproduced in a separate helicity-frame construction)** — the register node `eft_operator_basis` itself remains tier `to-derive` pending its own graduation screen; the frame-level result is banked in its boundary_condition (source: `calc/RESULTS_operator_basis.md`; `provenance/claims.json`).

Three honesty notes travel with this result and are constitutive of how the framework must be read:

(i) The TT restriction is chosen, not forced. Restricting $c_0 = 0$ — the pure-TT projector — is exactly the `p_tt_ansatz` choice: an input, not a consequence. The five-angle interrogation of 2026-08-02 (full Lorentz-covariant level, Barnes–Rivers 6-dimensional kernel space, core computation replicated in exact rational arithmetic by three independent verifiers) returned the same verdict from a strictly stronger level of generality: diffeomorphism invariance gets you to transverse and stops exactly one condition short of transverse-traceless. The decisive exhibited fact: linearized Einstein–Hilbert is *itself* $(1/2)k^2[P^{(2)} - 2P^{(0s)}]$ — GR's own response kernel carries a scalar component twice its spin-2 one, so no general principle can force a gravitational response kernel to be TT-only.

> **STATUS: ASSUMPTION (STRUCTURAL-SELECTION — CHOSEN, unanimous five-angle interrogation; not forced)** — canonical status for the TT-only projector (source: `provenance/claims.json` node `p_tt_ansatz`, boundary_condition).

(ii) The two-survivor exhaustiveness is scope-corrected. The 2026-08-14 Ward-scope correction, from the SCDP primary-literature read the register's own fence had flagged as owed (arXiv:2507.03103): dissipative and noise operators necessarily break the doubled diffeomorphism group to its diagonal, so imposing the *advanced-branch* identity is a symmetry no dissipative completion sustains; under the surviving diagonal identity the Ward constraint buys transversality on the retarded slot only, and the admissible open-gravity space is strictly larger. The correction *strengthens* the CHOSEN verdict (a larger space makes TT-only more of a choice) and surfaced the noise kernel's transversality as an input — subsequently discharged as a family-conditional theorem on rung1's fourth priced input (the 2026-08-17 owner ruling: the Ward-sourced zero plus N-positivity close the admissible (K_R, N) pair on the projector pair; `calc/noise_transversality_check.py`, recorded in `S_IF.md` §3).

(iii) The gauge footing is asymmetric. $P^{(2)}$ is invariant under the full linearized 4d diffeomorphism group (helicity protection); $P^{(0^s)}$ is invariant only under the spatial subgroup at fixed slicing — its full 4d-covariant status requires the Bardeen completion, which remains frontier-reserved. Every uniqueness statement in this section carries that frame/order fence.

II.3 · The admission gates

A candidate kernel pair (K_R, N) must pass four gates before it is admissible. Each gate has a declared status; none of them is a GRUT discovery, and the record says so.

Causality / analyticity. The coefficient functions c_i(ω) must be analytic in the upper half ω-plane — the retarded condition. Kramers–Kronig then links the elastic (storage) part Re χ to the dissipative part Im χ; the record's rung4 recovers the worldline-EFT tidal-response (Love-number) structure for the vacuum in exactly this standard linear-response sense.

> **STATUS: RECOVERED (borrowed Kubo/KK linear-response structure on the declared inputs) —** register node `rung4_love_kk`, tier shown; its GW-dissipation observable was computed and landed real-but-unobservable, ~21–62 orders below LIGO phase sensitivity (outcome B, closed) (source: `provenance/claims.json` node `rung4_love_kk`).

Retardation itself — *why* the kernel is retarded rather than advanced — is not derived:

> **STATUS: ASSUMPTION (EMPIRICAL-INPUT + STRUCTURAL — observed time-asymmetry; the audits located its origin in a past-boundary condition in every formulation examined) —** source: `GRUT_MODEL_FRAMEWORK.md` §2; the surviving unexplained input is one relative datum, the half-line/KMS alignment: **ASSUMPTION (one relative datum: the half-line/KMS alignment; five dressings audited, every closure consumed it)** (canonical status; Book V treats this front in full).

The KMS/FDT lock. In equilibrium the noise kernel is locked to the dissipation by the fluctuation–dissipation theorem, N tied to Im χ by the coth(ħω/2kT) factor, and KMS detailed balance is enforced as a *hard admission gate*: any candidate (χ, N) pair whose residual |G_K − coth·(G_R − G_A)| exceeds tolerance fails and is barred from the foundation (`gate/kms.py`).

> **STATUS: ASSUMPTION (borrowed standard identity, enforced as a hard admission gate; rung2) —** canonical status (source: `provenance/claims.json` node `rung2_kms_gate`).

The temperature in that lock carries zero freedom on the declared background:

> **STATUS: DERIVED (within declarations: forced by Hadamard/KMS on the declared background) —** $T_{dS} = H/2\pi$, canonical status; the register's only "forced given the background" credit (source: `provenance/claims.json` node `rung2_kms_gate`, ledger_note).

A corroborating computed record exists at curved order: Gate-E verified the FDT/KMS lock per H-order — PASS at $O(H^0)$, $O(H^1)$ (both sides identically zero, each computed) and $O(H^2)$, as a support identity within the declared validity domain $\omega \gg H$, with the coth → sgn grading *derived*, not asserted (the dS temperature is non-perturbative in the H grading). This is a W-0 record: computed and reported, unbanked (`PHYSICS_LEDGER/GATE_E_H2_FDT_KMS_AUDIT.md`).

Passivity, per channel. For any kernel decomposed on mutually orthogonal idempotent projectors, the matrix-sense passivity condition ($\omega \cdot \mathrm{Im}\, K$ positive-semidefinite) is *exactly equivalent* to the per-channel scalar conditions $\omega \cdot \mathrm{Im}\, c_i(\omega) \geq 0$, each channel independently, pointwise — with **no cross-channel rescue**: a violating channel's negative eigenvalue is unmoved by any amplification of a compliant channel. The KMS lock, carrying a common scalar thermal factor, is channel-diagonal at the same level, so a dissipation-sign violation survives into the noise rather than being masked.

> **STATUS: DERIVED (frame-free mathematics; PSD factorizes over orthogonal idempotents; machine-verified with a pre-registered four-mutant battery) —** register node `passivity_channel_diagonal`, tier shown, sealed prereg before the calc existed (source: `calc/RESULTS_x_no_pin.md`; `provenance/prereg/PREREG_X_NO_PIN_2026-08-09.txt`).

A guard travels verbatim with this lemma and is quoted here because downstream prose has historically been tempted to violate it: passivity "*can propagate a channel's vanishing … but can never source one. Any argument deriving channel annihilation from passivity alone is a category error and dies.*" The passivity floor is not a licence for the pure-TT ansatz and not evidence against it.

Positivity of N. The noise kernel is a positive-semidefinite covariance. The record's 2026-08-17 owner decomposition ruled this *constitutive* of what the framework already banks: the symmetrized correlator of any genuine state is PSD by construction, given the declared system/bath split and Gaussian truncation.

> **STATUS: ASSUMPTION (constitutive of the declared Gaussian-bath inputs; not a separate dial) —** source: `provenance/claims.json` node `rung1_inin_formalism`, ledger_note (owner ruling 2026-08-17).

II.4 · The two anti-overclaim theorems

Two theorem-grade structural facts about the admissible family are load-bearing precisely because they bound what the framework may claim of itself.

u1 — form-universality. Any local, causal open quantum system coarse-grains to the $S_{IF} = K_R + (i/2)N$ form. The framework is therefore a genuine universal IR *language* — and for exactly that reason the form confers no GRUT-specific content. GRUT adopts a universal language; it does not own the universality, and u1's truth must never be leveraged as support for kernel-universality (u2), which is a separate, open question (§II.8).

> **STATUS: RECOVERED (generic; u1: the form confers no GRUT-specific content)** — canonical status (source: `provenance/claims.json` node `u1_form_universality`).

x_no_pin — the action carries a family, not a point. Applying the channel-diagonal passivity lemma to the Ward-surviving two-channel family: every admissible kernel obeys the per-channel sign floors $\omega \cdot \mathrm{Im}\, c_2 \geq 0$ and $\omega \cdot \mathrm{Im}\, c_0 \geq 0$, pointwise — **and nothing more**. The admissible set is a convex cone, closed under independent nonnegative rescaling of each channel (verified through amplitude 10^6) and realizing *every* nonnegative ratio c_0/c_2. Passivity orients each channel, never bounds an amplitude, never selects a ratio. Route R3 closes as *classifier, not pinner*.

> **STATUS: DERIVED (enumerated frame/order; register tier `derived–pending`, pending exactly its hypothesis's open frontier — the operator basis's 4d-covariant completion)** — source: `provenance/claims.json` node `x_no_pin_theorem`; `calc/RESULTS_x_no_pin.md`.

Under the written action's declared normalization $x := c_0/c_0\hat{}$ (trace-only) (`S_IF.md` §5), the floor restates as x_diss(ω) ≥ 0 pointwise — a constraint on a *function*, never a bare inequality on a number. The transfer from the dissipative floor to the static/quasi-static modulus that observables couple to is *not* automatic: unconditional transfer is refuted by an exhibited passive counterexample (a negative contact term is invisible to passivity, causality, and the two-point KMS lock), and conditional transfer is derived at the class-level criterion $\chi\_\infty \geq 0$ (sufficient, not necessary).

> **STATUS: UNRESOLVED (register tier derived-pending; the conditional-transfer theorem is derived — criterion χ_∞ ≥ 0, sufficient not necessary — but the transfer question gates on the open sign of χ_∞; unconditional transfer REFUTED by passive counterexample)** — harmonized with Book IV §8.1 at corpus audit; register node `kk_static_transfer`, tier `derived–pending`; the residual sign of χ_∞ is a contact/renormalization datum in rung3's domain (source: `provenance/claims.json` node `kk_static_transfer`; `S_IF.md` §7).

The hand-chosen point of the family is then booked exactly as what it is:

> **STATUS: ASSUMPTION (STRUCTURAL-SELECTION; x_no_pin: the action carries a family, pins nothing)** — the x = 0 scalar-to-tensor choice, canonical status (source: `provenance/claims.json` nodes `p_tt_ansatz`, `x_no_pin_theorem`).

The consequence of the cone structure deserves stating in this book because it is a *mathematical* fact about the constitutive framework: the framework's commitment is a class; a class has no scale; the admissible set is an amplitude-homogeneous cone — every route from this framework to a number runs outside it. That is the derived structural reason the program's PREDICTED set is empty.

> **STATUS: EMPTY (nothing has earned entry; Book IX governs entry)** — the PREDICTED set, canonical status (source: `GRUT_PROGRAM_FREEZE.md` §3).

II.5 · Memory: the Mori–Zwanzig leg and the state of the finite-memory bet

The Mori–Zwanzig / Nakajima–Zwanzig projection formalism — slow variable, projected fast bath, memory kernel, generalized Langevin equation — is one of the framework's four standard-machinery legs, verified against the primary literature (Mori 1965; Zwanzig 2001; the register's source list). As with u1, the leg is borrowed: the NZ equation is an identity, its kernel is defined by the projector P, and MZ coarse-graining is many-to-one — distinct microscopics wash to the same kernel. The framework's *content* claim was always about the shape of the kernel, and that claim's record is the most heavily audited in the corpus.

The conjecture. The strongest GRUT form: the vacuum's memory kernel is single-pole (Debye) — finite memory, one relaxation time.

> **STATUS: UNRESOLVED (anchor-class, derived-pending; pole-vs-cut open; the Tier-4 computation found a CUT, not a pole, at flat scope)** — canonical status for the single-pole/finite-memory kernel (source: `provenance/claims.json` node `rung3_single_pole`).

The audited state of the conjecture, stated on its face. The record on this node is a model lesson in what ruthless tier-marking looks like, and this book reports it rather than smoothing it:

- *The phrase "the Mori–Zwanzig kernel" in the node's own statement does not denote a unique object.* `calc/mz_inheritance.py` (2026-08-19) showed the two standard candidates answer the key inheritance question oppositely: the symmetrized correlation carries the KMS factor's full Matsubara pole ladder, while the Kubo–Mori (canonical) correlation replaces coth by $2/(\beta\omega)$ and has no ladder at all; the generalized-Langevin friction kernel contains no temperature. The node's own finite-T history identifies its kernel as the symmetrized one, so the adverse reading applies. Recorded on the node without tier or ledger move — the tier is already the honest one.
- *At free level on the static patch, the single rate the conjecture names does not exist.* The free graviton bath's memory is a family indexed by multipole: the slowest surviving Matsubara rung is $(l+1)H$ (the graviton has no $l = 0, 1$), and at fixed l the surviving rungs are the whole infinite tower $n \geq l+1$. An exact sum rule ($\Sigma_l (2l+1)A_l = -\omega \sinh^2 x$, verified to 26+ digits) shows the l-summed local spectral density is exactly Ohmic with no rung zeros — the rung structure survives into the graviton bath solely because two partial waves are absent. The owner's final reconciliation: *the splitting mechanism is genuinely de Sitter* ($\beta = 2\pi/H$ places the split centrifugal zeros exactly on the ladder); *its survival in the summed bath is kinematic.* All recorded 2026-08-19 on the node, no tier move (source: `provenance/claims.json` node `rung3_single_pole`, tier_note; `calc/mz_inheritance.py`, `calc/finite_T_pole_structure.py`).

The reversal, part of the model's history and not a footnote. The finite-memory bet's sole in-house quantitative support (the tt_worldline decay) was reversed by corrected analysis — a real bug validated only at its blind point — and the prior ("something surely forces finite memory") was removed by enumeration: all seven candidate memory mechanisms deliver either no memory or unbounded, scale-free memory.

> **STATUS: ASSUMPTION, with REVERSED history on its face (the in-house "no memory time" computation was reversed; exact dS free-field theory forces infinite scale-free memory) —** finite memory time, canonical status (source: `GRUT_PROGRAM_FREEZE.md` §4; `RAI_GORILLA_T1.md` XVI-H).

What exact de Sitter actually forces: the constant tail. The massless minimally coupled field on dS_4 — hence each free TT graviton polarization — has an exact constant zero mode ($\Delta_- = 0$), and its retarded response keeps an **exactly constant tail $H^2/4\pi$ filling the interior of the light cone**, verified by a two-sided causality gate sited away from any degenerate point; conformal coupling gaps it at $m_{eff}^2 = 2H^2$.

> **STATUS: DERIVED (exact dS; gapped only at conformal coupling) —** canonical status; the surviving fixed point of every deletion test (source: `RAI_GORILLA_T1.md` XVI-G; `GRUT_PROGRAM_FREEZE.md` §3).

Whether that persistence survives interaction is the open computation the freeze names as reopening key O2: the interacting graviton zero-mode. The record's own mutation control shows the zero is unprotected against generic perturbation (an interaction-induced $m_{eff}^2 = 0.1H^2$ returns a finite rate ≈ $0.034H$ through the verified Starobinsky–Yokoyama channel); *lifted* fells the persistence claim, *protected* strengthens it materially.

> **STATUS: UNRESOLVED (O2 undone; decides the fixed point's referent) —** source: `GRUT_PROGRAM_FREEZE.md` §5; `RAI_GORILLA_T1.md` XVI-N.

II.6 · The derived kernel: the Tier-4 contract object

Against that backdrop of chosen structure and open memory, the program computed one kernel end-to-end at a declared scope — the contract-level retarded TT kernel, built through the validated chain vertex (T1) → massless TT bath (T2) → one-loop assembly (T3) → retarded completion (T4):

```
K_R(ω) = Σ_R(ω) = −(3/1280π²) ω⁴ L + H²·(−(13/480π²) ω² L) + [real local slot],
L = log(μ²/ω²) + iπ,    with H¹ = 0 identically.
```

Unconditional content: a branch point at $\omega = 0$ with a real-axis cut (the gapless two-graviton continuum); the frozen absorptive coefficients; retarded analyticity of the $+i\pi$ completion. Conditional content, banked *as* conditional: no additional real-axis zero of the resummed denominator on the reference slice only; resummed/first-order agreement iff $|\lambda| \ll 1$. **No pole claim is made.** Validity $\omega \gg H$ with $\varepsilon_H = (104/9)H^2/\omega^2$; $\omega \lesssim H$ is refused by the evaluator itself. The local slot is carried symbolically: at H^0, $c0 = c2 = 0$ exact by the Option-β D5 execution with c4 represented by the RG-invariant Λ_R, retained symbolic — one unresolved renormalization input; the H^2 locals c0′, c2′ are unresolved and fork-gated (no IR scale introduced).

> **STATUS: DERIVED (flat contract scope, ω ≫ H) —** the Tier-4 TT kernel with $H^1 = 0$ identically, canonical status; register node `kr_contract_retarded_tier4` banked 2026-09-01 as a scoped computed record at ledger 0 — *no derivation credit accrues to any GRUT rung* (source: `PHYSICS_LEDGER/WALL_KR_CONTRACT_RETARDED_VERDICT.md`; `provenance/claims.json` node `kr_contract_retarded_tier4`).

The benchmark consequence of that kernel, computed under pre-registered rules (`PHYSICS_LEDGER/WALL_KR_CONTRACT_BENCHMARK_VERDICT.md`, W-0):

```
Im χ(ω) = (3/1280π) ω⁴ + (13/480π) H²ω²  > 0
J_eff(ω) = (3/1280π) ω⁵ + (13/480π) H²ω³
```

The flat slice is a pure power law — log-slope of Im χ exactly 4 at every ω — i.e. **s = 5** in the registered J-convention, firmly convergent (Re $\chi(0) = 3/(2560\pi^2)$ exactly, no cutoff anywhere in the instrument). The registered s = 3 is *rejected by the register's own tolerance* (slope residual 2.0 ≫ TOL_S = 0.30); the s = 3 shape re-enters only as the curvature-induced $O(H^2)$ component, with an H^2-proportional coefficient the registered family excludes. The mechanism finding adjudicates the register's own rung3 derivation: s = 3 came from DOS ~ ω^2 with an assumed coupling weight; the actual TT-TT-TT vertex is two-derivative, contributing ω^4 in $|V|^2$ on the gapless cut.

> **STATUS: DERIVED (flat scope; rejects the framework's own registered s = 3)** — the spectral law s = 5, canonical status (source: `PHYSICS_LEDGER/WALL_KR_CONTRACT_BENCHMARK_VERDICT.md`).

Axis 2 of the benchmark (relaxational vs resonant) is INDETERMINATE with the missing component precisely named — the D5 local conditions; the registered test is scheme-hostage (the reference-slice sign crossing sits exactly at $\omega = \mu$ and moves with μ). The scheme-robust substatement: for every real local choice there is no resonance-class feature in-domain — Im χ monotone positive, no pole, no peak. Two boundaries of this whole section remain the owner's: the D5 renormalization conditions, and the kernel-vs-dressed object identity (the dressed reading would land Im $\chi \to$ const, the opposite side of the pre-registered convergence boundary).

> **STATUS: UNRESOLVED (owner fork, still owed: kernel vs dressed-propagator identity of the registered object; axis-2 verdicts must be re-run if the dressed reading is ruled)** — source: `PHYSICS_LEDGER/WALL_KR_CONTRACT_BENCHMARK_VERDICT.md` (sensitivity disclosure); `GRUT_PROGRAM_FREEZE.md` §3 UNRESOLVED.

A separate W-0 record (NOISE-A) confines the $\alpha = -2$ noise-sector behavior to the equal-time/secular mode-sum class, which no registered observable consumes; the finite-frequency noise kernel of the registered observable is an exact polynomial (`PHYSICS_LEDGER/WALL_KR_NOISE_IR_AUDIT.md`, unbanked).

II.7 · The $H^1 = 0$ theorem and the even-degree ladder class (Theorem II)

The Tier-4 kernel's most striking structural feature — the identically vanishing O(H) sector — was made the object of a nine-phase campaign (2026-09, frozen 2026-09-04) that decomposed, derived, and then honestly bounded it. The final object:

```
H¹ = F_state + F_weight + F_ladder + F_R ≡ 0   (pointwise, pre-angular, frozen construc
```

Four channels, four distinct cancellation depths, no channel promoted beyond its support:

- **S (state)** — the O(H) state term of the Bunch–Davies pair, killed at identity level: the canonical FRW Wronskian a^2W = const forces the amplitude direction (a standard identity); the declared BD prescription excludes the mixing direction (an input). DERIVED (amplitude) + DECLARED INPUT (mixing).
- **W (weight)** — the $(-2+2)(u+u')$ conformal-weight cancellation per key: the total is derived from the T1/T2 chains; the 2-vs-2 split is convention. DERIVED (total) + CONVENTION (split).
- **L (ladder)** — $\Lambda\_N = 0$ per transposition orbit: even momentum degree ⇒ reflection bridge ⇒ graded Gram symmetry ⇒ the antisymmetric weight annihilates. DERIVED (class-level) — this is the leg that generalizes.
- **R (remainder)** — the u-free vertex-grading remainder: exactly zero in the frozen construction, gated across all four cache configurations, mechanism *characterized* (frequency-insertion antisymmetry) but **underived**. CLOSED-AS-GATED.

The ladder leg's generalization is the campaign's — and arguably the program's — one genuinely new structural identity, **Theorem II**: for *any* entry family of even total momentum degree, any entry count, any symmetric slot pairing, under the fixed-ω reflection convention, $\Lambda\_N \equiv 0$ as a polynomial identity. Its ancestry is exactly two standard mathematical identities; support is gated at complete bases of degrees {0, 2, 4}, with odd counterexamples at {1, 3} and higher even degrees following by the parity identity (a derivation, not a gate). The boundary was located precisely: the true premise is the *evenness* of the momentum degree — a generic cosmological constant cannot break the ladder; odd degree is the obstruction. Einstein–Hilbert membership implies even-degree entries (tensor-level gate, 7,560 terms), so EH guarantees the ladder premise everywhere the construction reaches — but the EH-general statement is explicitly *not* established (probe-direction coverage, S/W at EH-general level, and the R channel are the named missing bridge).

> **STATUS: DERIVED (the one genuinely new structural identity; carries no confirmatory weight for GRUT)** — the $H^1 = 0$ four-channel cancellation + even-degree ladder class (Theorem II), canonical status (source: `PHYSICS_LEDGER/WALL_KR_H1_PHASE9_CLOSURE.md`; `provenance/claims.json` via the frozen phase artifacts).

The closure memorandum's binding interpretation is reproduced here because every book of this corpus must obey it: **no GRUT-specific premise occurs in the H^1 ancestry** (a machine-checked leaf audit over a hand-encoded 21-node graph — a search verdict, not a proof of absence), so GRUT was never under test, and $H^1 = 0$ must not be cited as evidence for GRUT in any downstream record. That no-citation rule is PROPOSED, awaiting owner ratification; until then it binds the record's authors and this corpus's default behavior. The deflationary consequence is stated without cushioning: the campaign isolated a computed identity *of the standard construction* at this order and scope.

II.8 · Kernel selection: derived from declared inputs, constrained but not selected

Does anything in the registered material *select* the Tier-4 kernel out of the admissible space of §II.2–II.3? The kernel-selection audit (2026-09-03, W-0, 23/23 battery) answered with a six-way decomposition of the ω^4 claim: dimensional analysis alone forces nothing (the registered ω^3 comparator was dimensionally admissible); the ω^4 weight is forced by the two-derivative vertex — *and the vertex order is input microphysics (Einstein–Hilbert), not a principle*; the gapless two-graviton cut comes from the declared massless bath; scale-freeness at $H = 0$ then forces the exponent; one loop computes the coefficient. Five admissible-but-different kernel families were exhibited without computing anything (the Λ_R family; the (c0′, c2′) family; different bath content; a gapped bath; a different admissible state) — **every alternative is excluded by an input — vertex, bath, state, scheme — never by a principle.** On the forced hierarchy, the GRUT kernel sits at "derived from microscopic dynamics *given the declared inputs*" and at "constrained — not uniquely selected — within the admissible space."

> **STATUS: UNRESOLVED (the selection question is open — novelty class B; no registered GRUT-specific principle reduces the admissible kernel space beyond standard constraints; W-0 audit record, unbanked)** — source: `PHYSICS_LEDGER/WALL_KR_KERNEL_SELECTION.md`.

The one candidate with selective power if proven is u2 — kernel-universality across microscopic/quantum-gravity completions:

> **STATUS: UNRESOLVED (to-derive, default-BROKEN; likely under-determined out of the gate; graduation requires ≥ 2 distinct completions flowing to the same IR-kernel class)** — source: `provenance/claims.json` node `u2_kernel_universality`.

II.9 · State descriptions and effective dynamics downstream

States. The framework's state-side declarations are the D1–D5 sheet: the BD-analogue state (D3), scheme/no-IR-scale (D5), order of limits (D1), gauge (D2/D4) — each a priced structural selection (`GRUT_PROGRAM_FREEZE.md` §3). The BD declaration is not decorative: in Theorem II's decomposition it is *the entire reason the S channel's mixing direction vanishes* — a concrete instance of a state declaration doing constitutive work. For the equilibrium statements the state is KMS at the background temperature; and at the algebraic level the record carries the audited caveat that the canonical tensor-product system/bath split does not exist in the relevant algebra type (III_1) — what survives translation is a one-sided inclusion, with the crossed-product construction relocating rather than discharging the state-selection input.

> **STATUS: RECOVERED (borrowed; audited verdict B-INPUT-RELOCATION: the unexplained input moves, it is not discharged)** — Type $III_1 \to II_1$ via the CLPW crossed product, canonical status; Book VIII treats this front (source: `GRUT_MODEL_FRAMEWORK.md` §2).

Effective dynamics. Integrating out the bath yields the reduced master equation; the unitary core is Schrödinger; the noise kernel supplies decoherence selecting a pointer basis.

> **STATUS: RECOVERED (standard decoherence machinery on declared inputs)** — pointer-basis selection via the noise kernel, canonical status; Book III treats the quantum sector, including the Born-measure door GRUT does not open (source: `provenance/claims.json` node `rung6_qm_limit`; `GRUT_MODEL_FRAMEWORK.md` §5).

At zero memory the framework collapses to GR:

> **STATUS: RECOVERED (largely by identity; KERNEL-STANDARD, scoped)** — GR at zero-memory collapse, canonical status: compatibility, not correspondence evidence (source: `GRUT_PROGRAM_FREEZE.md` §3).

One class-level dynamical exclusion survives with real discriminating force, and its non-ownership is part of its statement: every purely relaxational kernel — Debye, multi-pole, Cole–Cole — stays on one side of $w = -1$; only an oscillatory pole pair crosses. Any observed crossing falsifies the whole passive class at once, GRUT included.

> **STATUS: DERIVED (class-level; explicitly not GRUT-specific)** — the kernel-class discriminator, canonical status (source: `PHYSICS_LEDGER/RUNG7_TWO_POLE_COMPARISON.md`; `NO_GO_LEDGER.md` entry 3).

Finally, the RRT arm contributed the program's cleanest theorem-grade product, a structural statement about response itself: intervention response is reducible for *all* linear dynamics — unitary, dissipative, non-invertible, Lindblad alike — escaped only by nonlinearity.

> **STATUS: DERIVED (RRT arm: intervention response reducible for all linear dynamics; escaped only by nonlinearity)** — canonical status (source: `GRUT_PROGRAM_FREEZE.md` §3).

II.10 · The absence map

Where the constitutive record is silent, this book says so. The following have **no GRUT account in the record** or are explicitly fenced open:

1. **The bath's microscopic content.** S_IF is defined at the level of its kernels; the bath Hilbert space is integrated out and *not specified* — rung3's frontier, priced where it lives. The transport self-energy Σ from the bath's internal dynamics (the object that would decide pole-vs-cut at the memory level) is frontier-reserved: specify and hand out, do not approximate in-house.
2. **The $\omega \lesssim H$ regime.** UNASKABLE at current declarations, on four obstructions — not false; the evaluator refuses it (source: `GRUT_PROGRAM_FREEZE.md` §3; `PHYSICS_LEDGER/WALL_KR_CONTRACT_RETARDED_VERDICT.md`).
3. **The D5 local conditions and Λ_R.** One unresolved renormalization input, owner-ruled symbolic; the H^2 locals fork-gated. Until chosen, axis-2 of the benchmark cannot be adjudicated and no scheme-invariant Re χ sign statement exists.
4. **Amplitudes and scales.** No amplitude theorem is executed; every verdict in §II.6 is invariant under overall positive rescaling; the cone structure of §II.4 is why.
5. **The 4d-covariant / Bardeen / Riegert–Paneitz completion of the operator basis** — frontier-reserved; every frame-level uniqueness statement carries this fence. The parity-odd sector is outside the enumeration entirely.
6. **An in-house causality ceiling (Israel–Stewart-type).** Not computable in-house — it needs a relaxation time, sound speed, and entropy density named nowhere in the corpus; reclassified as a rung3-dispatch sub-question. No number was computed.

7. **A GRUT-specific selection principle for the kernel** — none found (§II.8); and no GRUT gate principle was found for the H^1 identity (§II.7). Both absences are recorded findings, not oversights.

> **STATUS: UNMAPPED / UNRESOLVED (as itemized; each absence carries its source above)** — an absence map is valid content; nothing here is bridged by invention.

Sources drawn from

- `books/CORPUS_CHARTER.md` (status vocabulary; canonical status table)
- `GRUT_MODEL_FRAMEWORK.md` (authoritative model presentation)
- `GRUT_PROGRAM_FREEZE.md` (stopping rule; consolidated ledger)
- `provenance/claims.json` (register nodes: `rung1_inin_formalism`, `rung1_ontology_finite_memory`, `rung2_kms_gate`, `rung3_single_pole`, `rung4_love_kk`, `p_tt_ansatz`, `eft_operator_basis`, `passivity_channel_diagonal`, `x_no_pin_theorem`, `kk_static_transfer`, `u1_form_universality`, `u2_kernel_universality`, `kr_contract_retarded_tier4`)
- `S_IF.md` (the formal action specification)
- `calc/RESULTS_operator_basis.md`
- `calc/RESULTS_x_no_pin.md`
- `PHYSICS_LEDGER/WALL_KR_KERNEL_SELECTION.md`
- `PHYSICS_LEDGER/WALL_KR_CONTRACT_RETARDED_VERDICT.md`
- `PHYSICS_LEDGER/WALL_KR_CONTRACT_BENCHMARK_VERDICT.md`
- `PHYSICS_LEDGER/WALL_KR_H1_PHASE9_CLOSURE.md`
- `PHYSICS_LEDGER/GATE_E_H2_FDT_KMS_AUDIT.md`
- `PHYSICS_LEDGER/WALL_KR_NOISE_IR_AUDIT.md`
- `NO_GO_LEDGER.md`
- `RAI_GORILLA_T1.md` (the dS tail / zero-mode complex; the memory-mechanism enumeration)
- `RAI_DIALECTIC_CHAMBER.md` (the tail's three-round stability; W-0 chamber record)
- `GRUT_V1_PLAIN.md` (the four standard-machinery legs)

External literature cited only where the record already cites it: arXiv:2507.03103 (the SCDP Ward-scope correction); Feynman–Vernon 1963, Caldeira–Leggett 1983, Mori 1965, Zwanzig 2001, Callen–Welton 1951, Kubo 1966 (via register `sources` keys).

Gaps in this book

1. **The Mori–Zwanzig leg has no dedicated committed synthesis document**; its state is reconstructed here from the rung3 node's tier_note history and the cited calc files

(`calc/mz_inheritance.py`, `calc/finite_T_pole_structure.py`). A reader wanting the full 2026-08-19 sequence must read the node text itself.

2. **The exact dS tail's primary derivation artifact** is cited via `RAI_GORILLA_T1.md` and the freeze rather than a single dedicated calc RESULTS file; the underlying computations (static-patch response, zero-mode complex) are spread across session records this book did not re-verify.
3. **The Wall-A A3-x finite-masters chain** (the dimensional-regularization master integrals feeding the Tier-3 loop) is not narrated here; this book picks up the chain at the frozen T1–T4 verdicts.
4. **The noise-sector fork** ($\omega \lesssim H$, white-floor regime, owner-held) is named but not developed — it is outside every declared scope this book covers.
5. **Book cross-references are prospective**: Books III (quantum sector), V (KMS/arrow), VI (cosmology), VIII (algebraic/relational) are cited by scope, and were not available to check for consistency at the time of writing. (Corpus audit 2026-09-06: the landed books were checked against these citations and found consistent.)
6. **The eft_operator_basis graduation screen** (which would move the node past `to-derive`) has not run; if it runs, §II.2's status blocks must be re-read against its outcome.
7. **The H^1 no-citation rule is PROPOSED, not ratified**; this book obeys it as default behavior. If the owner rules otherwise, §II.7's framing sentence changes, though no status changes.